

Li-Yun (James) Wang

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Google Scholar: [Publication Link](#)

Bio

Dedicated professional with expertise in computer vision, specializing in image and video analysis and image synthesis. Proven experience in the practical applications of Generative Adversarial Networks (GANs), Machine Learning, and Deep Learning algorithms. Currently engaged in a collaborative project focused on artist image style measurement and transfer via vision language and generative models. Enthusiastic about exploring and contributing to the forefront of research in computer vision through machine and deep learning techniques.

Selected Publication

- A. Ghildyal, **L. Wang**, and F. Liu. WP-CLIP: Leveraging CLIP to Predict Wolflin's Principles in Visual Art. Submitted it to ICCVW 2025
- A. Althoupety, **L. Wang**, W. Feng, and B. Rekabdar. Night-to-Day: Unpaired Image-to-Image Translation for Nighttime Pedestrian Detection. European Conference on Artificial Intelligence (ECAI), July 2024
- A. Althoupety, **L. Wang**, W. Feng, and B. Rekabdar. DaFF: Dual Attentive Feature Fusion for Multispectral Pedestrian Detection. Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), June 2024
- A. Althoupety, **L. Wang**, W. Feng, and B. Rekabdar. Illuminating the Bias in Pedestrian Detection. IEEE International Symposium on Multimedia (ISM), December 2023
- L. Wang** and W. Feng. Video as Text: A New Paradigm for Flexible Video Analysis. IEEE International Symposium on Multimedia (ISM), December 2023
- L. Wang**, A. Rhodes, W. Feng. Class Specialized Knowledge Distillation. The 16 th Asian Conference on Computer Vision (ACCV), 2022
- A. Saqlain, **L. Wang**, and Z. Liu. SL-CycleGAN: Blind Motion Deblurring in Cycles using Sparse Learning. IEEE International Conference on Cloud Computing and Big Data Analytics (ICCCBDA), 2022
- L. Wang** and Z. Akhtar. CCAP: Cooperative Context Aware Pruning for Neural Network Model Compression. IEEE International Symposium on Multimedia (ISM), December 2021
- L. Wang**. Adversarial Perturbation Suppression using Adaptive Gaussian Smoothing and Color Reduction. IEEE International Symposium on Multimedia (ISM), December 2021
- L. Wang**, Y. Jalalpour, and W. Feng. Context-Aware Image Denoising with Auto-Threshold Canny Edge Detection to Suppress Adversarial Perturbation. arXiv, 2021
- Y. Jalalpour, **L. Wang**, W. Feng, and F. Liu. FID: Frame Interpolation and DCT-based Video Compression. IEEE International Symposium on Multimedia (ISM), December 2020
- Y. Jalalpour*, **L. Wang***, R. Feng, and W. Feng. Leveraging Image Processing Techniques to Thwart Adversarial Attacks in Image Classification. IEEE International Symposium on Multimedia (ISM), December 2019
- L. Chen, **L. Wang**, and C. Su. Human Detection in Surveillance Video. International Journal of Pattern Recognition and Artificial Intelligence. Vo.28, No. 2, March 2014
- L. Chen, P. Lee, **L. Wang**, and Mark Liao. Integration of Multiple Shape Features for Human Detection in Videos. IHH-MSP, 2014
- L. Chen, Y. Hung, and **L. Wang**. An Integrated Approach to Image Retrieval. International Conference on Telecommunications and Signal Processing (TSP), 2012
- L. Wang**. An Integrated Approach to Human Detection. Master Thesis. Fu Jen Catholic University, 2012
- L. Chen, H. Hsu, **L. Wang**, and C. Su. Violence Detection in Movies. International Conference on Computer Graphics, Imaging, and Visualization (CGIV), 2011

Work Experience

AI/ML Engineer

January 2025 – Present

Tiposi Inc.

Milpitas, CA

- **Technical Skills:** Python, PyTorch, Computer Vision, Generative Models, and Problem Solving
- Researched and developed an ensemble method for stroke detection from microwave signals in Python and PyTorch, and demonstrated a significant **5 - 6 %** improvement in Accuracy and Recall.
- Researched and developed a two-stage image generation framework for MRI generation from microwave signal inputs in Python and PyTorch, and demonstrated a significant **5 - 6 dB** and **1 - 2** improvement in PSNR and SSIM.
- Collaborated with cross-background teammates to improve data consistency across hardware devices, ensuring seamless interaction between devices.

Software Engineering Intern - Video Codec Hardware

April 2022 – June 2022

Apple Inc.

Cupertino, CA

- **Technical Skills:** C++, Python, Pytorch, Computer Science, Git, Teamwork, and Problem Solving.
- Developed concise and insightful summaries, demonstrating a keen understanding of complex concepts and trends.
- Took an active role in the research and development of innovative video coding algorithms. Designed a recurrent neural network with LSTM layers in Python and PyTorch for accelerating the process of encoding micro-blocks during video encoding for DCT-based video codec frameworks. The new framework improves video encoding time with a **50%** reduction.
- Led the prototyping phase, experimenting with various approaches to enhance video compression and quality.
- Utilized Git for version control and managed workflow processes, ensuring team synchronization and project efficiency. Successfully implemented a new video codec framework.

Software Engineering Intern - Machine Learning Engineering

May 2021 – August 2021

Hewlett-Packard (HP) Inc.

Spring, TX

- **Technical Skills:** Python, Git, Bash, Pytorch, OpenCV, and Model Training in CUDA
- Prototyped an innovative heart-rate monitoring system with a lightweight human face detection module in Python, PyTorch, and OpenCV, conducting system performance evaluations to enhance system use-ability and maintain the accuracy of heart-rate changes in Colab, Numpy, and OpenCV. Optimized the detection module through skipping techniques and achieved a remarkable **3X improvement** in inference time.
- Optimized the prototyped monitoring system, improving inference efficiency and maintaining heart-rate monitoring quality by analyzing and refining the monitoring system through replacing the detection module with a face segmentation module in the monitoring system, achieving a remarkable **7X improvement** in processing speed.
- Optimized the segmentation component in Python and PyTorch, improving storage usage with a **50%** reduction and maintaining segmentation performance by re-designing the segmentation component.
- Collaborated with cross-background teammates to integrate the heart-rate monitoring system feature into laptop prototypes, ensuring seamless interaction between software components.
- Utilized Git for version control and managed workflow processes, ensuring team synchronization and project efficiency. Successfully implemented multiple versions of the system.

Software Engineering Intern - Bixby Data Pipeline

October 2019 – April 2020

Samsung Research America

Mountain View, CA

- **Technical Skills:** Python, Pyspark, AWS S3 and EC2, MinIO, Kafka, Git, MongoDB, GCP, Kibana, and Elasticsearch
- Implemented an innovative audio data pipeline in Python, AWS, MongoDB, integrating a keyword detection model into the pipeline for training data generation and conducting the generation process within a tight **4-day** timeframe. Visualized statistics of audio data from the results of Elasticsearch in Kibana.
- Optimized the pipeline in Python and Multi-thread APIs, improving processing time by refining data downloading from AWS S3 and MinIO and keyword segmentation components and achieving a remarkable **2X improvement** in processing speed.
- Implemented an efficient end-to-end generic audio keyword extraction pipeline in Python, PySpark, and MongoDB, utilizing a **learning-based audio keyword detection model** to detect certain keywords in audio data from GCP. This optimized pipeline processes millions of audio data entries daily, reducing the processing time by **50%** and enhancing overall data throughput.
- Utilized Git for version control and managed workflow processes, ensuring team synchronization and project efficiency. Successfully integrated new features of the pipeline into existing pipelines.

Graduate Student Researcher

September 2016 – December 2023

Portland State University

Portland, OR

- **Technical Skills:** Machine and Deep Learning, Python, Pytorch, C/C++, FFmpeg, Generative Model, OpenCV, Foundational Model, and HPC.
- **DaFF: Dual Attentive Feature Fusion for Multispectral Pedestrian Detection:** (1) Cleaned and processed diverse datasets, including daytime and nighttime variations, to prepare for training Faster R-CNN and YOLO models. (2) Researched and introduced a novel **dual-fusion technique** for enhanced multi-modal feature fusion. (3) demonstrated a significant **2%** improvement in average precision via experiments and analysis. (4) The work was accepted for a CVPR workshop 2024.
- **Night-to-Day: Unpaired Image-to-Image Translation for Nighttime Pedestrian Detection:** (1) Cleaned and processed diverse datasets, including daytime and nighttime variations, to prepare data for one- and two-stage object detection models and training of generative models. (2) Researched and introduced a lightweight and unpaired **GAN-based model** for the conversation about daytime images given dark or nighttime images. (3) demonstrated a **1-2%** improvement in average precision through experiments and analysis. (4) The work was accepted for ECAI (European Conference on Artificial Intelligence) 2024 (Acceptance Rate = 23%).

- **Video as Text: A New Paradigm for Flexible Video Analysis:** (1) Led the development of 'Video as Text: A New Paradigm for Flexible Video Analysis,' a software framework for streamlining content-based video analysis. (2) Designed modular components with generic inputs/outputs using middleware concepts to enhance flexibility. (3) Introduced an adaptive naming mechanism for dynamic storage and reuse of intermediate results. Achieved a 57.14 ms object detection speed on 640x480 resolution video with 8900 frames, demonstrating the efficiency of the framework. (4) The work was accepted for ISM (International Symposium on Multimedia) 2023 (Acceptance Rate = 24.69%).
- **Learning-Based Image Deblurring:** (1) Led the development of a novel image deblurring technique using a **CycleGAN-based framework**, incorporating Perceptual Loss and Sparse Learning for enhanced robustness. (2) Collaborated with the team to refine the approach, demonstrating a significant **1-2 dB** improvement in PSNR and SSIM metrics through experiments and analysis.
- **Class Specialized Knowledge Distillation:** (1) Led research in model compression for cross-domain applications, developing the **Class Specialized Knowledge Distillation (CSKD)** framework to create compact, high-performance specialized neural networks. (2) Validated compression techniques through extensive testing, achieving an optimal balance between model size and performance across various scenarios. (3) demonstrated a significant **5-6%** improvement in classification accuracy through experiments and analysis. (4) The work was accepted for the ACCV (Asian Conference on Computer Vision) 2022 (Acceptance Rate = 33%).

Side Research Project

Artistic Style Similarity Assessment June 2024 - February 2025

Technical Skills: Python, Pytorch, Vision Language Model, HPC, and OpenAI APIs

- Researched and developed a novel VLM (Vision-Language Model)-based artistic style assessment metric for artistic image-style assessment problems in Python, OpenAI CLIP, and PyTorch.
- Conducted experimentation and successfully predicted Wolfflin’s five principles in seconds by the proposed VLM framework.
- Collaboratively wrote a manuscript and submitted it to an ICCV workshop 2025.

Text-to-Image Diffusion Model Evaluation for Cultural Awareness February 2025 - Present

Technical Skills: Python, Pytorch, Vision Language Model, and HPC

- Researching and designing evaluation protocols for cultural awareness evaluation in text-to-image diffusion models in Python, Pytorch, Diffuser, OpenRouter, and Huggingface.
- Applying Python, Pytorch, GenAI, Computer Vision, Deep learning to model bias evaluation in text-to-image diffusion models.

Education

Portland State University Sep. 2016 – Mar. 2024

Ph.D. in Computer Science, GPA: 3.65 Portland, OR, USA

Portland State University Mar. 2015 – Jun. 2017

MS in Computer Science, GPA: 3.59 Portland, OR, USA

Fu-Jen Catholic University Sep. 2010 – Jun. 2012

MS in Computer Science, GPA: 4.00 New Taipei, Taiwan

Fu-Jen Catholic University Sep. 2006 – Jun. 2010

BS in Computer Science, GPA: 3.80 New Taipei, Taiwan